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RV COLLEGE OF ENGINEERING®
(An Autonomous Institution affiliated to VTU)
VII Semester B. E. Examinations March -2022

Computer Science and Engineering

PARALLEL ARCHITECTURE AND DISTRIBUTED PROGRAMMING

Time: 03 Hours

Maximum Marks: 100

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2, 7 and 8 are compulsory. Answer any one full question from 3 and 4 & one full question from 5 and 6.

PART-A

1	1.1	A vector processor can handle strides greater than one, called _____.	01
	1.2	Define critical path of a task dependency graph.	01
	1.3	Compare write invalidate and write update protocols.	02
	1.4	Relaxing the W->W ordering yields a _____ model.	01
	1.5	Define chaining.	02
	1.6	Analyze sequential consistency.	02
	1.7	The following loop has multiple types of dependencies. Find only the true data dependencies. <pre> for (i = 0; i < 100; i = i + 1) { Y[i] = X[i]/c; /*S1*/ X[i] = X[i]+c; /*S2*/ Z[i] = Y[i]+c; /*S3*/ Y[i] = c - Y[i]; /*S1*/ } </pre>	02
	1.8	What is chunk scheduling? Identify the potential problem with chunk scheduling.	02
	1.9	Identify the use of MPI_Test and MPI_Wait function with respect to non-blocking communication operations	02
	1.10	Use the GCD test to determine whether dependencies exist in the following loop: <pre> for (i = 0; i < 100; i = i + 1) { X[2 * i + 3] = X[2 * i] * 5.0; } </pre>	02
	1.11	MPI provides a general method for partitioning a graph of processes known as _____.	01

1.12	Consider a loop: $\text{for } (i = 0; i < 100; i = i + 1)$ $A[i + 1] = A[i] + C[i]; \text{ /*S1*/}$ $B[i + 1] = B[i] + A[i + 1]; \text{ /*S2*/}$ $\}$ Assume that A, B and C are distinct, non-overlapping arrays. What are the data dependencies among the statements S1 and S2 in the loop?	02
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PART-B

2	a	Illustrate multiprocessor cache coherence problem with an example. Briefly explain the basic schemes for enforcing coherence.	08
	b	Explain the Directory based Cache Coherence Protocol indicating all the state transition diagram for an individual cache block.	08
3	a	Describe the basic structure of a vector architecture, VMIPS consisting of all the primary components	06
	b	Identify the significance of Vector Length Register and Vector Mask Register.	06
	c	Briefly explain conditional branching in GPUs.	04
		OR	
4	a	Show the code for MIPS and VMIPS for the DAXPY loop. Assume that the starting addresses of X and Y are in Rx and Ry respectively.	06
	b	Illustrate the several innovations introduced in Fermi to bring GPUs much closer to mainstream system processors.	06
	c	Suppose we have 8 memory banks with a bank busy time of 6 clocks and a total memory latency of 12 cycles. How long will it take to complete a 64-element vector load with a stride of 1? With a stride of 32.	04
5	a	Classify the various decomposition techniques and explain any two of them in detail.	08
	b	Examine the methods for containing or reducing the interaction overheads among concurrent tasks.	08
		OR	
6	a	Illustrate the different characteristics of the tasks have a large influence on the suitability of a mapping scheme.	08
	b	Discuss the Data-Parallel model and the Pipeline or Producer-Consumer Model.	08
7	a	Give the prototype and the use of the following collective communication operations of MPI: i. MPI_Gather ii. MPI_Reduce iii. MPI_Bcast iv. MPI_Barrier	08

	b	Consider two matrices A and B of size $n \times m$. Let $q \times r$ be the number of processes in the grid arranged in q rows and r columns. Develop and MPI program for multiplication of A and B matrices on a $q \times r$ process grid using cannon's algorithm.	08
8	a	Illustrate various forms of parallelism allowed by OpenACC execution model.	06
	b	Analyze implementation of Pipelining across multiple devices by OpenACC.	06
	c	Compare OpenACC parallel and kernel constructs used for parallelizing the iterations of loop.	04